Annihilating Branching Brownian Motion

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BBM

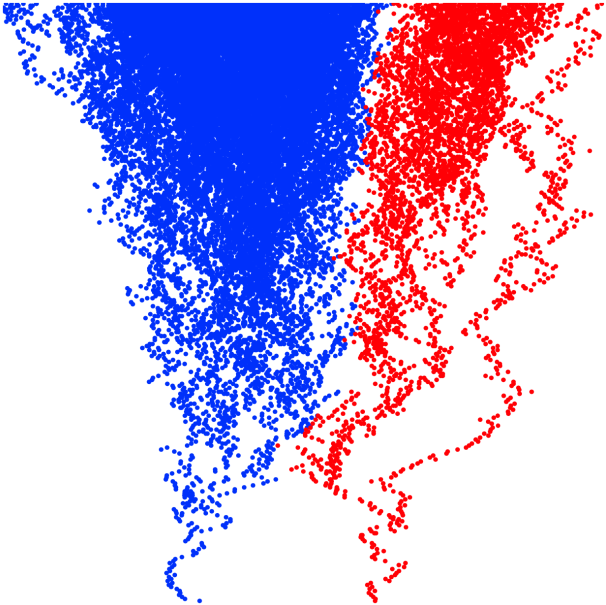
Branching Brownian motion

- Start with a single particle
- Particles follow Brownian motion on $\mathbb{R}$
- At rate 1, particles split into two

ABBM

Annihilating Branching Brownian motion

- Start with $\pm$ particles at ± 1
- $\pm$ particles give birth to $\pm$ particles
- The $\pm$ families evolve like BBMs,
  except that
  if a pair of $\pm$ particles meet, they **annihilate**



Coexistence

Theorem 1

- With positive probability,
  the $\pm$ families **coexist**

Interface

- Let $I^{\pm}(t)$ be the location of the left/rightmost $\pm$ particle at time t
- Define the **interface**

$$I(t) = \frac{I^{-}(t) + I^{+}(t)}{2}$$

to be the midpoint between the $\pm$ families

Linear interface

Theorem 2

- When the $\pm$ families coexist,
  the interface is **asymptotically linear**

- The limit
$$\lambda_* = \lim_{t \to \infty} \frac{I(t)}{t}$$
  exists almost surely
- Has full support on $[-\sqrt{2}, +\sqrt{2}]$
  and no atoms

Why linear?

Intuition #1

- At any t, equal number of $\pm$ particles have hit interface
- Consider an absorbing boundary $I(t) = \lambda t$
- Particles on either side hit the boundary at equal rates
- If $\lambda > 0$
  - **fewer** $+$ particles, but **many hit** the boundary
  - **more** $-$ particles, but **most do not hit** the boundary
- If $I(t)$ is far from linear, no longer true

Why linear?

Intuition #2

- **Reinforcement**, like in a Pólya urn
- After annihilation, space opens
- More likely filled by dominate family

Kesten

BBM with absorption

- **Kesten** (1978)
- Start BBM at $+1$
- Particles absorbed at 0
- Drift λ towards the origin
- $\lambda = \sqrt{2}$ is critical
- Consistent with Theorem 2

Many questions

Question #1

- **Competition** rather than annihilation
- When $\pm$ particles meet
  - $+$ wins with probability p
  - $-$ wins with probability $1-p$
- Coexistence for all $p \in (0,1)$?

Many questions

Question #2

- Start with a
  $+$ particle at 0 and
  $-$ particles at ± 1
- Are the interfaces linear?

Many questions

Question #3

- Start with a
  PPP(γ_+) of $+$ particles on $(0, \infty)$ and
  PPP(γ_-) of $-$ particles on $(-\infty, 0)$
- How does the interface behave?

Basic martingale

- Start BBM with 1 particle at 0
- At time t,
  expect $\mathbb{E}(N_t) = e^t$ many particles
- $(e^{-t}N_t, t \geq 0)$ is a non-negative martingale,
  (so converges almost surely)
- This is $\lambda = 0$ in the following martingale ...

Additive martingale

- For any λ, consider

$$Y_\lambda(t) = \sum_j e^{\lambda X_j(t)}$$

summing over locations $X_j(t)$ of particles at time t

- Then

$$W_\lambda(t) = e^{-\hat{\lambda} t} Y_\lambda(t), \qquad \hat{\lambda} = 1 + \frac{\lambda^2}{2}$$

is a non-negative martingale

Biggins

- Let
$$W_\lambda = \lim_{t\to\infty} W_\lambda(t)$$
denote the additive martingale limit

- **Biggins** (1977, 1992)
- For $|\lambda| < \sqrt{2}$,
convergence in L^1 and $\mathbb{P}(W_\lambda > 0) = 1$
- For $|\lambda| \geq \sqrt{2}$,
$\mathbb{P}(W_\lambda = 0) = 1$
- W_λ is analytic in $\lambda \in (-\sqrt{2}, \sqrt{2})$

Witness of activity

Key observation

- Main contribution to

$$W_\lambda(t) = e^{-\hat{\lambda} t} \sum_j e^{\lambda X_j(t)}$$

  is from particles **near** λt

- (Within, e.g., distance $t^{1/2+\epsilon}$)

Conservative coupling

- When a pair of $\pm$ particles meet, instead of annihilating, suppose they coalesce into a single **neutral** particle
- A neutral particle continues like a BBM (giving birth to more neutral particles) and does not interact with anything else

Key observation

- The $+$ and neutral particles form a BBM
- The $-$ and neutral particles form a BBM
- Consider $W_\lambda^\pm(t)$ for these two (dependent) BBMs

Proof of coexistence

- Fix $\lambda \in (0, \sqrt{2})$
- With positive probability, at some $t > 0$, there are exactly two $\pm$ particles at distance $2a$
- So let us assume that initially $\pm$ particles at $\pm a$, and show $\mathbb{P}(\text{coexist}) \geq 1/2$ from here
- Note that
$$W_\lambda^\pm \stackrel{d}{=} e^{\pm a} W_\lambda$$
- Therefore, for large a,
$$\mathbb{P}(W_\lambda^- < 1 < W_\lambda^+) \geq 3/4$$

Proof of coexistence

- On the event $W_\lambda^- < 1 < W_\lambda^+$, for large t,

$$W_\lambda^+(t) - W_\lambda^-(t) = e^{-\hat{\lambda} t}\Big(\sum_i e^{\lambda X_i^+(t)} - \sum_j e^{\lambda X_j^-(t)}\Big) > 0$$

  since the neutral particles cancel

- But this is only possible if there **exist** $+$ particles for all large t, so $\mathbb{P}(+ \text{ survives}) \geq 3/4$
- By symmetry, taking $\lambda \in (-\sqrt{2}, 0)$, we find that $\mathbb{P}(- \text{ survives}) \geq 3/4$
- Therefore $\mathbb{P}(\text{coexist}) \geq 1/2$

Proof of linear interface

- **Main step:** show $W_\lambda^- \neq W_\lambda^+$ for $-\sqrt{2} < \lambda < \sqrt{2}$
- To do this, couple two ABBM systems, where in one the first branching event is suppressed
- Difference is a BBM, so cannot have $W_\lambda^- = W_\lambda^+$ in both
- Total variation between them is $\leq 1/e$
- Hence $W_\lambda^- = W_\lambda^+$ in unsuppressed ABBM with probability at most $(1 + 1/e)/2 < 1$
- But then, by Lévy's 0–1 law, with probability 0

Thank you